

```
lokesh@cybercub:~$ sudo apt update
[sudo] password for lokesh:
Get:1 https://download.docker.com/linux/ubuntu jammy InRelease [48.8 kB]
Get:2 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
```

.

.

```
lokesh@cybercub:~$ sudo apt upgrade -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
```

.

.

Before installing QEMU KVM check yours cpu is supported for the hardware virtualization
Make sure yours hardware is greater than 0 then yours cpu support hardware virtualization if not
greater it cant support kvm

```
lokesh@cybercub:~$ egrep -c '(vmx|svm)' /proc/cpuinfo
16
```

Diagnostic to check whether yours server or computer can run hardware accelerated KVM
virtual machine it says whether yours system is KVM capable or not

Check if yours cpu supports hardware virtualization extensions like intel VT or amdv check if
these virtualization features are enabled in yours bios and UEFI settings outputs the message to
inform you whether yours system is kvm capable or not

```
lokesh@cybercub:~$ sudo apt install cpu-checker -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
```

msr-tools

.

.

```
Setting up cpu-checker (0.7-1.3build1) ...
Processing triggers for man-db (2.10.2-1) ...
```

```
lokesh@cybercub:~$ kvm-ok
INFO: /dev/kvm exists
KVM acceleration can be used
```

```
lokesh@cybercub:~$ sudo apt install qemu-kvm libvirt-clients libvirt-daemon-system bridge-utils
virtinst libvirt-daemon -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Note, selecting 'qemu-system-x86' instead of 'qemu-kvm'
bridge-utils is already the newest version (1.7-1ubuntu3).
bridge-utils set to manually installed.
.
.
Processing triggers for libc-bin (2.35-0ubuntu3.8) ...
Processing triggers for man-db (2.10.2-1) ...
```

The service will start automatically when the system boots up

```
lokesh@cybercub:~$ sudo systemctl enable --now libvirt
```

Check activity of virtualization service

```
lokesh@cybercub:~$ sudo systemctl status libvirt
```

● libvirtd.service - Virtualization daemon

Loaded: loaded (/lib/systemd/system/libvirtd.service; enabled; vendor preset: enabled)

Active: active (running) since Fri 2024-11-15 21:46:30 IST; 9min ago

TriggeredBy: ● libvirtd-admin.socket

● libvirtd.socket

● libvirtd-ro.socket

Docs: man:libvirtd(8)

<https://libvirt.org>

Main PID: 25977 (libvirtd)

Tasks: 21 (limit: 32768)

Memory: 9.0M

CPU: 423ms

CGroup: /system.slice/libvirtd.service

└─25977 /usr/sbin/libvirtd

└─26095 /usr/sbin/dnsmasq --conf-file=/var/lib/libvirt/dnsmasq/default.conf --leasefile-ro

--dhcp-script=/usr/lib/libvirt/libvirt_leaseshelper

└─26096 /usr/sbin/dnsmasq --conf-file=/var/lib/libvirt/dnsmasq/default.conf --leasefile-ro

--dhcp-script=/usr/lib/libvirt/libvirt_leaseshelper

Nov 15 21:46:30 cybercub systemd[1]: Started Virtualization daemon.

Nov 15 21:46:30 cybercub dnsmasq[26095]: started, version 2.90 cachesize 150

Graphical interface to manage yours virtual machine so install virtual manager ,it provide the user friendly to create modify run and connect to virtual machines behind the scenes it can interact with the various virtualization solutions like kvm qemu

You are setting up to manage the virtual machine by GUI

```
lokesh@cybercub:~$ sudo apt install virt-manager -y
```

```
Reading package lists... Done
```

```
Building dependency tree... Done
```

```
Reading state information... Done
```

```
.
```

```
.
```

```
Processing triggers for man-db (2.10.2-1) ...
```

```
Processing triggers for mailcap (3.70+nmu1ubuntu1) ...
```

To manager kvm without root privileges you need to add user to the virtualization group

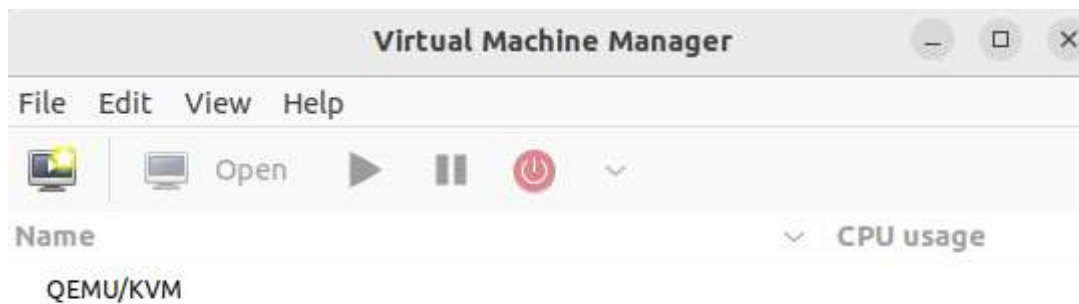
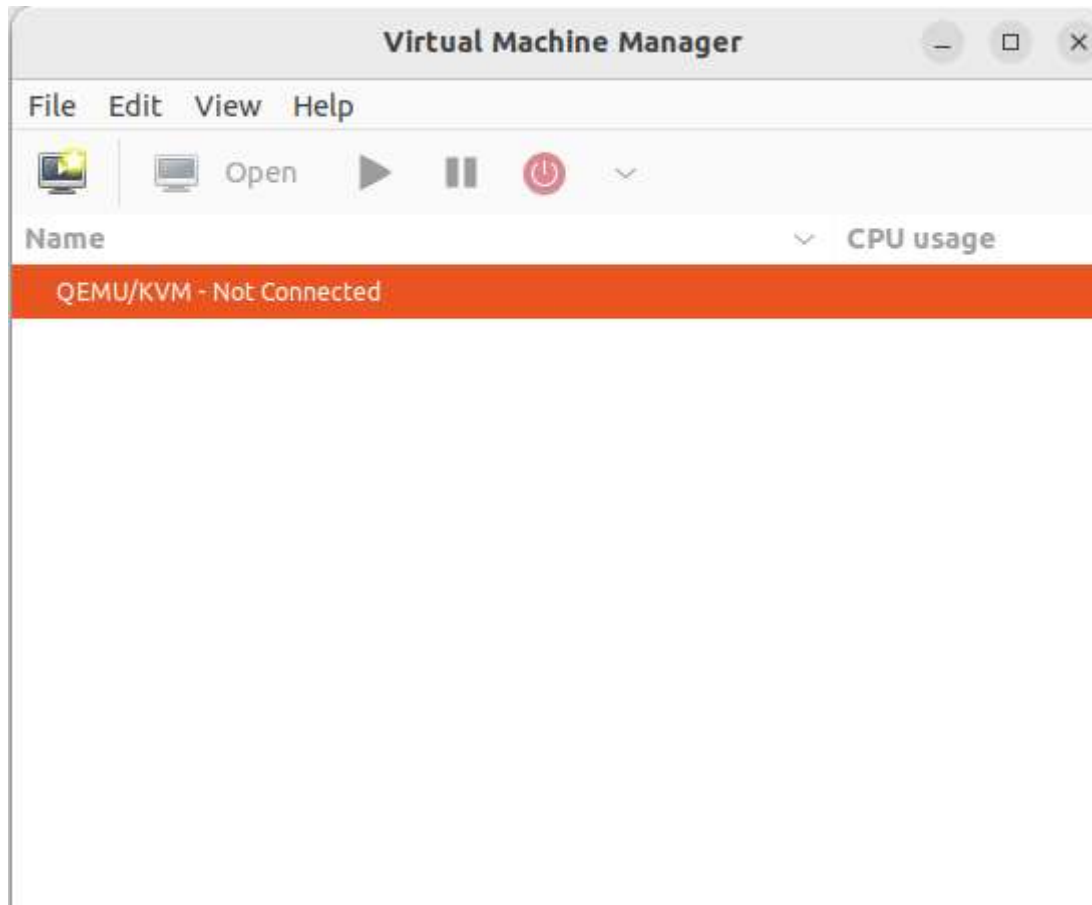
This command will add currently logged in user

```
lokesh@cybercub:~$ sudo usermod -aG kvm $USER
```

```
lokesh@cybercub:~$ sudo usermod -aG libvirt $USER
```



Restart the machine for the current user to have admin privileges to the virtualization group
You can also continue with root
lokes@cybercub:~\$ sudo virt-manager



New VM



Create a new virtual machine

Step 1 of 5

Connection: QEMU/KVM

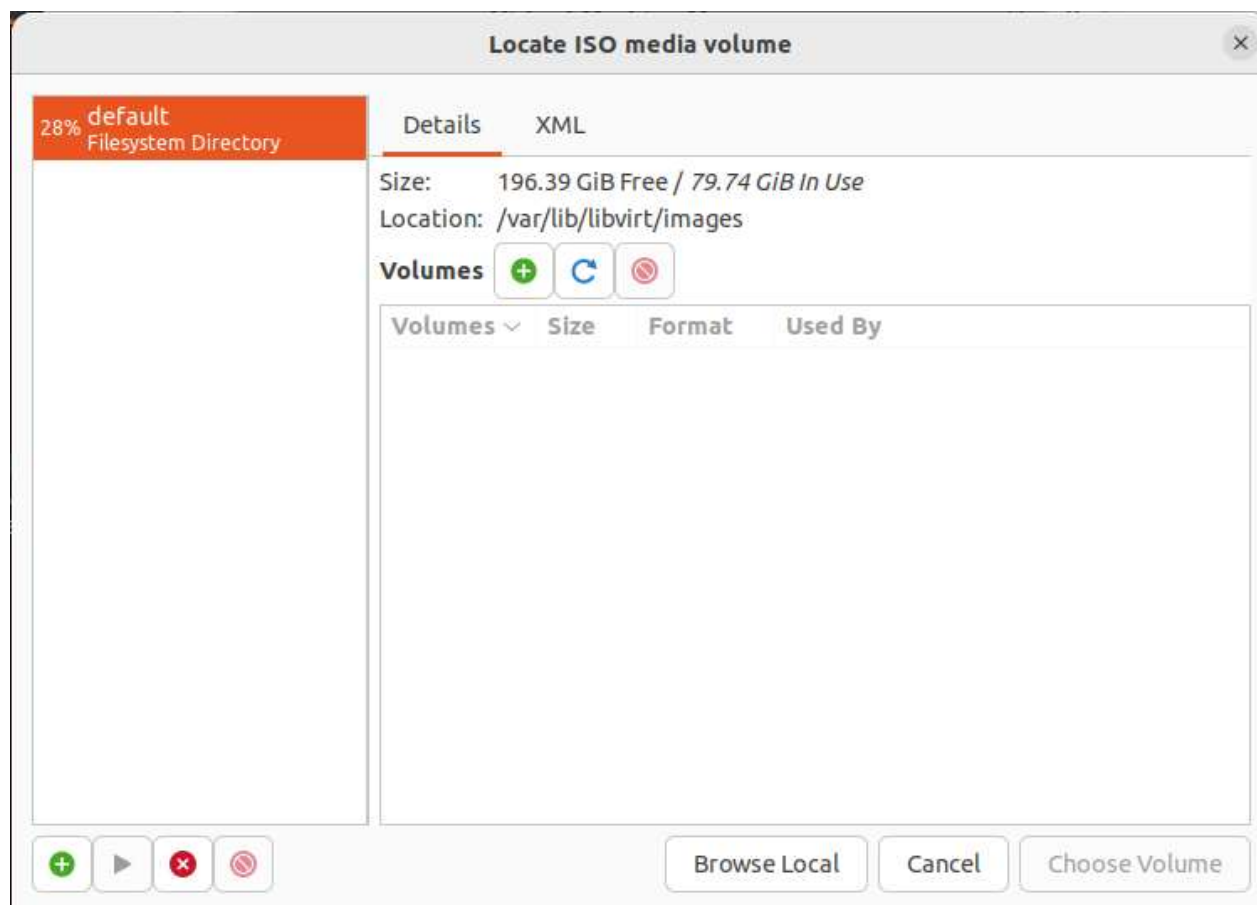
Choose how you would like to install the operating system

- ☒ Local install media (ISO image or CDROM)
- ☐ Network Install (HTTP, HTTPS, or FTP)
- ☐ Import existing disk image
- ☐ Manual install

Cancel

Back

Forward



New VM



Create a new virtual machine

Step 2 of 5

Choose ISO or CDROM install media:

Downloads/ubuntu-24.04.1-live-server-amd64.iso



Browse...

Choose the operating system you are installing:

Q Ubuntu 24.04 LTS



☒ Automatically detect from the installation media / source

Cancel

Back

Forward

New VM



Create a new virtual machine

Step 4 of 5

- ☒ Enable storage for this virtual machine
- ☒ Create a disk image for the virtual machine

40.0 — + GiB

196.4 GiB available in the default location

- ☐ Select or create custom storage

Manage...

Cancel

Back

Forward

New VM



Create a new virtual machine

Step 3 of 5

Choose Memory and CPU settings:

Memory:

Up to 15744 MiB available on the host

CPUs:

Up to 8 available

Cancel

Back

Forward

New VM



Create a new virtual machine

Step 5 of 5

Ready to begin the installation

Name: T-POT

OS: Ubuntu 24.04 LTS

Install: Local CDROM/ISO

Memory: 8000 MiB

CPUs: 2

Storage: 40.0 GiB /var/lib/libvirt/images/T-POT.qcow2

☒ Customize configuration before install

▼ Network selection

Virtual network 'default' : NAT ▼

Cancel

Back

Finish



Network configuration

[Help]

Configure at least one interface this server can use to talk to other machines, and which preferably provides sufficient access for updates.

NAME	TYPE	NOTES
[enp1s0	eth	- ▶]
static	192.168.29.10/24	
52:54:00:92:a8:99 / Red Hat, Inc. / Virtio 1.0 network device		
[Create bond ▶]		

Edit enp1s0 IPv4 configuration

IPv4 Method: [Manual ▼]

Subnet: 192.168.29.0/24

Address: 192.168.29.10

Gateway: 192.168.29.1

Name servers: 8.8.8.8
IP addresses, comma separatedSearch domains:
Domains, comma separated[Save]
[Cancel][Done]
[Back]

File Virtual Machine View Send Key



Profile configuration

[Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name:

Your servers name:

The name it uses when it talks to other computers.

Pick a username:

Choose a password:

Confirm your password:

[Done]